

Taut fillings

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Abstract

Sleator, Tarjan, and Thurston asked: given a triangulation σ of the 2-sphere, what is the minimum number of tetrahedra needed to extend σ to a triangulation of the ball? Call this minimum $\text{tetvol}(\sigma)$. Let X be the associated integral 2-cycle, and let $\text{Zvol}(\sigma)$ be the minimum L_1 -norm of an integral 3-chain M with $\partial M = X$. Certainly $\text{tetvol}(\sigma) \geq \text{Zvol}(\sigma)$. We show here that equality holds, and in fact any optimal filling M arises from an extension of σ to a shellable simplicial complex homeomorphic to the 3-ball. This complex is a flag complex: every clique in its 1-skeleton occurs as a simplex. The key to the proof is the general fact that any optimal filling of an integral n -cycle splits under disjoint union, connected sum, and more generally what we call almost disjoint union, where summands are supported on sets that overlap in at most $n + 1$ vertices.

1 Overview

Let $\Delta = \Delta(\Omega)$ be the $|\Omega| - 1$ -simplex with vertices Ω , viewed as the abstract simplicial complex consisting of all subsets of Ω . Let $C_n = C_n(\Omega) = C_n(\Delta(\Omega), \mathbf{Z})$ and $Z_n = Z_n(\Omega) = Z_n(\Delta(\Omega), \mathbf{Z})$ be its integral n -chains and n -cycles. A *filling* of an n -cycle $X \in Z_n$ is any $n+1$ -chain $M \in C_{n+1}$ with $\partial M = X$. (Here and throughout we'll assume $n \geq 1$.) Let $\text{Zvol}(X)$ be the minimum L_1 -norm of a filling of X . Call $M \in C_{n+1}$ *taut* if it is an optimal filling of its boundary, i.e., if $|M| = \text{Zvol}(\partial M)$,

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For $X \in C_n$ let $\text{Vert}(X)$ be the set of all the vertices of all the n -simplices to which X assigns non-0 weight. For any taut filling M of X we have $\text{Vert}(M) = \text{Vert}(X)$. (Cf. Proposition 4 below.) This is why we write C_n and Z_n without reference to Ω .

Call an n -cycle $X + Y \in Z_n$ an *almost disjoint union* if

$$|\text{Vert}(X) \cap \text{Vert}(Y)| \leq n + 1.$$

This notion generalizes both disjoint union and connected sum along an n -simplex. Ellison [2] showed the Zvol adds under almost disjoint union: $\text{Zvol}(X + Y) = \text{Zvol}(X) + \text{Zvol}(Y)$. This means that *some* taut filling M of X splits into a sum $M = M_X + M_Y$ of taut fillings of X, Y . Here we amplify Ellison's Corollary 2 to show (Theorem 1) that when $n \geq 2$, *any* taut filling of $X + Y$ splits.

In place of integral chains we can use chains with coefficients in $\mathbf{Q}$ or $\mathbf{R}$. Evidently $\text{Rvol} \leq \text{Qvol}$. But computing Rvol is a linear programming problem with rational coefficients, so in fact $\text{Qvol} = \text{Rvol}$. Ellison used LP duality to prove that Qvol adds under almost disjoint union. Now for $n \geq 2$ we get the stronger result (Corollary 1) that taut $\mathbf{Q}$ -fillings split: Multiplying a taut $\mathbf{Q}$ -filling M of X by a common denominator q yields a taut $\mathbf{Z}$ -filling qM of qX : Split the $\mathbf{Z}$ -filling qM and divide by q to get a splitting of the $\mathbf{Q}$ -filling M .

Now let σ be a simplicial triangulation of S^2 ; let $X(\sigma) \in Z_2(\Delta(\text{Vert}(\sigma)), \mathbf{Z})$ be either one of the 2-cycles that arises from σ by orienting its 2-simplices; and write $\text{Zvol}(\sigma) = \text{Zvol}(X(\sigma))$. Let $\text{tetvol}(\sigma)$ be the minimum number of 3-simplices required to extend σ to a simplicial triangulation τ of B^3 . We will show that

$$\text{Zvol}(\sigma) = \text{tetvol}(\sigma).$$

Certainly $\text{Zvol} \leq \text{tetvol}$, because any extension τ induces a filling of $X(\sigma)$. Theorem 2 states that any taut filling τ of $X(\sigma)$ arises in this way. As corollaries, we get that τ is a shellable flag complex: It can be assembled one simplex at a time, and any clique in its 1-skeleton occurs as a simplex. The proof of Theorem 2 relies on Theorem 1.

2 Background

Our interest in Zvol stems from the work of Sleator, Tarjan, and Thurston [5]. They observed that coning from a vertex of maximal degree shows that

a v -vertex triangulation σ of S^2 has $\text{tetvol} \leq 2v - 4 - \max\deg(\sigma)$. If $v \geq 13$ $\max\deg(\sigma) \geq 6$, so $\text{tetvol} \leq 2v - 10$. They produced examples for which $\text{tetvol} = 2v - 10$, provided v is larger than some unspecified bound, and conjectured that such examples should exist for any $v \geq 13$. Their approach was to realize σ as an ideal hyperbolic polyhedron with volume $2vV_0 - O(\log(v))$, where V_0 is the volume of an equilateral ideal tetrahedron, which is maximal. This implies $\text{tetvol} \geq 2v - O(\log(v))$. From there they bootstrapped their way up to showing that $\text{tetvol} = 2v - 10$. They suggested [5, p. 697] that in fact $\text{Qvol} = 2v - 10$, which would immediately imply $\text{tetvol} = 2v - 10$. Mathieu and Thurston [3] produced a class of examples, different from the examples of Sleator, Tarjan, and Thurston, for which they could show that $\text{Qvol} = 2v - 10$, still under the assumption that v is sufficiently large. In [1] we produced examples for any $v \geq 13$ with $\text{Qvol} = 2v - 10$, confirming the conjecture of Sleator, Tarjan, and Thurston. Theorem 2 here tells us that whenever we have $\text{Qvol} = 2v - 10$, or just $\text{Zvol} = 2v - 10$, any taut filling must arise from a triangulation of the ball.

About Qvol versus Zvol . In [1] we described triangulations of S^2 with $\text{Qvol} < \text{Zvol}$. In all such examples that we know of with $\max\deg \leq 6$, the gap between Qvol and Zvol is < 1 , so that $\lceil \text{Qvol} \rceil = \text{Zvol}$. Since Qvol and Zvol add under connected sum, the gap between them can be arbitrarily large (Ellison [2]), but taking the connected sum pushes $\max\deg$ above 6.

3 Taut fillings

We can think of an n -chain $X \in C_n$ as a multiset of non-cancelling oriented simplices:

$$X = \sum_{t \in X} t.$$

In this sum t denotes an oriented simplex

$$t = [x_0, \dots, x_n] = [x_{\pi(0)}, \dots, x_{\pi(n)}], \quad \pi \text{ an even permutation,}$$

and each unoriented simplex contributes a number of terms corresponding to its multiplicity.

The size of M as a multiset is its L_1 -norm $|M|$. Write $U \subset M$ if U is a sub-multiset of M . This happens just when $|M| = |U| + |M - U|$.

Proposition 1. *If M is taut and $U \subset M$ then U is taut.*

Proof. This is clear, but let's drag it out. If U is not taut, there is a smaller filling V of ∂U . Take $M' = M - U + V$. In terms of multisets, this means that we substitute V for U , and then do any required cancellation of oppositely oriented simplices of $M - U$ and V . We have

$$\partial M' = \partial M - \partial U + \partial V = \partial M$$

and

$$|M'| \leq |M - U| + |V| = |M| - |U| + |V| < |M|,$$

contradiction. $\square$

4 Coning

For $x \in \Omega$, $U \in C_n$ let

$$\text{nbhd}(x, U) = \sum_{t \in U: x \in \text{Vert}(t)} t,$$

$$\deg(x, U) = |\text{nbhd}(x, U)|,$$

$$\text{maxdeg}(U) = \max_x \deg(x, U).$$

The *cone from x to U* is the $(n+1)$ -chain

$$\text{cone}(x, U) = \sum_{t \in U} \text{adj}(x, t),$$

consisting of all the non-trivial oriented $(n+1)$ -simplices $\text{adj}(x, t) = [xx_0 \dots x_n]$ obtained by adjoining x to $t = [x_0 \dots x_n] \in U$. If $x \in \text{Vert}(t)$ then t doesn't contribute to the sum, so $|\text{cone}(x, U)| = |U| - \deg(x, U)$.

If U is closed then $\partial \text{cone}(x, U) = U$, so

Proposition 2. *For any $X \in Z_n$*

$$\text{Zvol}(X) \leq |X| - \text{maxdeg}(X) \quad \square$$

If $U \in Z_n$ and $x \notin \text{Vert}(U)$ then $\deg(x, U) = 0$ and $|\text{cone}(x, U)| = |U|$. In this case we call $\text{cone}(x, U)$ a *complete cone*. By Proposition 2 a non-trivial complete cone is not taut. (This is a long-winded way of saying that instead of coning from $x \notin U$, we could have coned from some $x \in U$.) Thus:

Proposition 3. *If M is taut it contains no non-trivial complete cone.* $\square$

Call $x \in \text{Vert}(M) \setminus \text{Vert}(\partial M)$ an *internal vertex* of M . If x is internal to M then $\text{nbhd}(x, M)$ is a complete cone, so:

Proposition 4. *If M is taut then it has no internal vertices.* $\square$

5 Almost disjoint unions

Recall that if $X, Y \in Z_n$ we call $X + Y$ an almost disjoint union if

$$|\text{Vert}(X) \cap \text{Vert}(Y)| \leq n + 1.$$

The most interesting special case is a connected sum, where $t = [c_0, c_1, \dots, c_n]$ occurs once in X and $-t = [c_1, c_0, \dots, c_n]$ occurs once in Y . For example, if $n = 1$ with X a cycle of length p and Y a cycle of length q the connected sum $X + Y$ is a cycle of length $p + q - 2$. Here Zvol adds, because $\text{Zvol}(X) = p - 2$, $\text{Zvol}(Y) = q - 2$, and

$$\text{Zvol}(X + Y) = (p + q - 2) - 2 = \text{Zvol}(X) + \text{Zvol}(Y).$$

But not every filling of $X + Y$ splits, because a taut filling may contain any 2-simplex with vertices in $\text{Vert}(X + Y)$.

We want to show that fillings do split when $n \geq 2$.

For $A \subset \Omega$, $p \in A$ define

$$\begin{aligned} \pi_{A,p} : \Omega &\rightarrow A \\ \pi_{A,p}(x) &= x \text{ if } x \in A \text{ else } p \end{aligned}$$

and let

$$K_*(A, p) : C_*(\Omega) \rightarrow C_*(A)$$

be the induced chain map. This is a projection of $C_*(\Omega)$ onto $C_*(A)$.

Let A, B be finite vertex sets sharing the vertices $C = A \cap B$, and let $c = |C|$. Observe that $C_c(C)$ and $Z_{c-1}(C)$ are trivial, because $\Delta(C)$ contains no c -simplex, and (up to orientation) only one $c-1$ -simplex, with nothing to cancel its boundary.

Let

$$C_*(A, B) = C_*(A) \oplus C_*(B).$$

For $p, q \in C$ define the mapping

$$g_*(p, q) = K_*(A, p) \oplus K_*(B, q) : C_*(A \cup B) \rightarrow C_*(A, B).$$

Proposition 5. *Let $(X, Y) \in C_n(A, B)$. If $|A \cap B| \leq n$ we can recover (X, Y) from $X + Y$. If $(X, Y) \in Z_n(A, B)$ this holds also when $|A \cap B| = n + 1$.*

Proof. Let $C = A \cap B$. We can assume $|C| \geq 1$. (Add a brand new point to C if necessary.) We claim that for any $p, q \in C$ (not necessarily distinct) we have

$$g_n(p, q)(X + Y) = (X, Y).$$

Indeed,

$$K_n(A, p)(X + Y) = X + K_n(A, p)(Y).$$

The second term belongs to $C_n(C)$, which is trivial when $|C| \leq n$. If $Y \in Z_n(B)$ the second term belongs to $Z_n(C)$, which is trivial when $|C| \leq n + 1$. $\square$

Theorem 1. *If $|A \cap B| \leq n + 1$, for all $(X, Y) \in Z_n(A, B)$ we have*

$$\text{Zvol}(X + Y) = \text{Zvol}(X) + \text{Zvol}(Y).$$

And as long as $n \geq 2$, for any $M \in \text{taut}(X + Y)$ we have $M = M_X + M_Y$ with $M_X \in \text{taut}(X)$, $M_Y \in \text{taut}(Y)$.

Note. Ellison [2] proved the additivity of Zvol . What's new here is the splitting of taut fillings when $n \geq 2$. This result resembles Theorem 6.2 of Pournin and Wang [4] about flip paths. Like theirs, our proof uses a variation on the normalization technique of Sleator, Tarjan, and Thurston [5, Lemma 7]. It would be nice to fit these results under one roof.

Proof. Again let $C = A \cap B$. We can assume $|C| = n + 1$, as this is the hardest case. And we might as well go ahead and take $n = 2$, $|C| = 3$, as this case illustrates all the issues.

Take any $M \in \text{taut}(X + Y) \subset C_3(A \cup B)$. Pick distinct points $p, q \in C$, and let

$$(M_X, M_Y) = g_3(p, q)(M).$$

(For now we'll suppress the dependence of M_X, M_Y on p, q .)

Because $g_*(p, q)$ is a chain map we have $\partial_3 M_X = X$, $\partial_3 M_Y = Y$:

$$(\partial_3 M_X, \partial_3 M_Y) = \partial_3(g_3(p, q)(M)) = g_2(p, q)(\partial_3 M) = g_2(p, q)(X + Y) = (X, Y).$$

We want to show that under the map $g_3(p, q)$ any $t \in M$ dies either in M_X or M_Y , because then we'll get additivity of Zvol from

$$\text{Zvol}(X + Y) = |M| \geq |M_X| + |M_Y| \geq \text{Zvol}(X) + \text{Zvol}(Y) \geq \text{Zvol}(X + Y).$$

Let's call a 3-simplex a 'tet' for short. Say that a tet $t \in M$ has type $CCXY$ if $t = \pm[c_1c_2x_1y_1]$ for $c_1, c_2 \in C$, $x_1 \in A \setminus C$, $y_1 \in B \setminus C$. Similarly for types $XXXX$, $CXXX$, $CXXY$, $XXXY$, etc. The first two are pure X cases, meaning that they live in $C_3(A)$; the last two are hybrid cases.

Any $XX..$ tet dies in M_Y ; any $YY..$ tet dies in M_X .

The remaining cases to check are the pure cases $CCCX$, $CCCY$, and the hybrid case $CCXY$. A $CCCX$ tet $t = \pm[c_1c_2c_3x_1]$ dies in M_Y because $K_3(B, q)(t) = [c_1c_2c_3q]$, which vanishes because $q \in C = \{c_1, c_2, c_3\}$ produces a repeated vertex. Likewise a $CCCY$ tet dies in M_X . The more interesting case is $CCXY$: The key is that since $|C| = 3$, $\{c_1, c_2\}$ cannot be disjoint from $\{p, q\}$, so it dies on one side or the other.

So

$$|M| = |M_X(p, q)| + |M_Y(p, q)|,$$

and Zvol adds. Note how we're now emphasizing the possible dependence of M_X, M_Y on p, q . When $n = 1$ the choice of p, q can indeed make a difference: We can get a different pair (M_X, M_Y) if we switch p and q . (Think about the connected sum of two cycle graphs.)

But when $n \geq 2$ we will now show that all tets are pure X or pure Y , so M_X consists of all the pure X tets, and ditto for M_Y . This will make $M = M_X + M_Y$ with $M_X \in \text{taut}(X)$, $M_Y \in \text{taut}(Y)$.

Again our test case is $n = 2$.

The key observation is that, now that we know that every tet dies on one side or the other, we know that no tet can die on both sides, because that would make $|M| > \text{Zvol}(X) + \text{Zvol}(Y)$.

An $XXYY$ tet would die on both sides, so there can be none of these.

Any $pqXY$, $pXXY$, or $qXYY$ would die on both sides, and p, q are arbitrary, so this rules out all $CCXY, CXXY, CXYY$.

The only remaining hybrids are $XXXY$ and $XYYY$. Let's rule out $XXXY$, leaving $XYYY$ to symmetry.

Fix any $y_0 \in B \setminus C$, and suppose there is some $XXXy_0$ tet in M . Let $U \subset M$ consist of all such $XXXy_0$ tets. We claim that $y_0 \notin \text{Vert}(\partial U)$. For let s be any 2-simplex of the form XXy_0 , the only kind of 2-simplex containing y_0 that could belong to ∂U . Any $t \in M$ of which s or $-s$ is a face must be a hybrid tet, and the only possibility is $XXXy_0$, so $t \in U$. Because $\partial M = 0$ the signed multiplicity of s vanishes in ∂M , hence also in ∂U , because the tets of M with $\pm s$ as a face all belong to U . So $y_0 \notin \text{Vert}(\partial U)$, making U a complete cone on y_0 , contradicting 3. So there is no such $XXXy_0$ tet, ruling

out $XXXX$, and with it $YYYY$.

So there are no hybrid tets, meaning that M splits, as claimed. $\square$

The splitting of taut fillings depends on the assumption $n \geq 2$. Let's consider what goes wrong when $n = 1$. Here the hybrid types are CXY , XXY , XYX . Neither pXY nor qXY dies on both sides, so we can't rule out CXY . Failing that, the XXy_0 tets needn't form a complete cone, because $[x_0x_1y_0]$ can continue across $[x_1y_0]$ to $-[px_1y_0]$, so we can't rule out XXY either.

The foregoing proof works equally well over $\mathbf{Q}$, providing we permit ourselves to work with fractional multisets. Alternatively, we can clear denominators as in the introduction above. Either way, we have:

Corollary 1. *Qvol adds under almost disjoint union, and for $n \geq 2$ taut $\mathbf{Q}$ -fillings split.* $\square$

6 Filling a triangulation of the 2-sphere

Theorem 2. *Let σ be a simplicial triangulation of S^2 , and let M be a taut filling of σ . Then M is clean, and arises from a simplicial triangulation of B^3 .*

Proof.

Call (σ, M) a *filling pair* if σ is a simplicial triangulation of S^2 with M a taut filling of σ . Call a filling pair *good* if M is clean and arises from a simplicial triangulation of B^3 . We aim to show that all filling pairs are good.

We will argue by contradiction. Let (σ, M) be a bad filling pair for which $|M|$ is minimal over all filling pairs. Let $v, e = 3v - 6, f = 2v - 4$ count vertices, edges, and faces of σ . Obviously $v > 4$. Because taut fillings split under connected sum, we can assume that σ is prime (not a connected sum along a triangle), and in particular (this is all we will need) that there is no vertex of degree 3.

Call a tet $t \in M$ *eligible* if it shares two of its oriented faces with σ . (It can't share more than two faces, since σ has no vertex of degree 3.) By Proposition 2

$$|M| = \text{Zvol}(\sigma) \leq f - \max\deg(\sigma).$$

We need the usual supply of disjointly eligible tets. Since M is a coefficiented chain, we phrase the count at the level of tets themselves, not coefficient

copies. For each tet u occurring in M , let $A(u)$ be set of faces of σ which are oriented faces of u . Thus $|A(u)| \leq 2$. Every face of σ occurs in at least one of the sets $A(u)$, because its coefficient in ∂M is nonzero and has the sign specified by σ .

Now list the tets of M and reveal the sets $A(u)$ one at a time, keeping track of the faces of σ already covered. A new tet contributes 0, 1, or 2 previously uncovered faces. If r tets contribute two new faces, then

$$f \leq |M| + r.$$

Since $|M| \leq f - \max \deg(\sigma)$, it follows that $r \geq \max \deg(\sigma)$. The tets which contribute two new faces give at least $\max \deg(\sigma)$ disjointly eligible tets, meaning eligible tets whose pairs of boundary faces are mutually disjoint. For the nonce we will only need two such tets.

Consider an eligible tet t with boundary faces $s_1 = [a, b, c]$, $s_2 = [b, a, d]$. Removing t from M produces a filling $M - t$ of a 2-complex σ_t , where the faces s_1, s_2 have been replaced by faces $[c, d, b]$, $[d, c, a]$, with the unoriented edge ab having been ‘flipped’ to become edge cd . There are two possibilities for the simplicial complex σ_t :

- (1) Edge cd was not an edge of σ ; σ_t is a triangulation of S^2 .
- (2) Edge cd was an edge of σ ; σ_t is the almost disjoint union of two triangulations σ_1, σ_2 of S^2 , conjoined along the edge cd .

In case (1), by minimality of $|M|$ $(\sigma_t, M - t)$ is a good filling pair, so $M - t$ is clean and arises from a triangulation of B^3 . The same will be true of $M = (M - t) + t$, providing that adding the tet t back in doesn’t produce a non-simplicial triangulation. That would happen just if t occurs in $M - t$, which is to say that t occurs with multiplicity 2 in M . Here’s where we use the fact that we can choose an eligible tet u whose boundary faces are disjoint from those of t . (Really we only need that it not be a clone of t , it could have one of its boundary faces in common with t .) In this case $(\sigma_u, M - u)$ is a bad filling pair, because the presence of $2t$ in $M - u$ means that $M - u$ does not arise from a simplicial triangulation, whether of B^3 or anything else. So $2t$ doesn’t belong to M ; $M = (M - t) + t$ arises from a triangulation of B^3 ; (σ, M) is a good filling pair; and we have a contradiction.

Case (2) is similar. $M - t$ splits into $M_1 + M_2$, with filling pairs (σ_1, M_1) and (σ_2, M_2) , both good by minimality of $|M|$. So both M_1 and M_2 arise from simplicial triangulations of B^3 , and M arises from the complex obtained by gluing these triangulations together with t between them. Again the only

possible obstruction is that t was already present in $M - t$; if so, remove a second eligible tet u instead, as above, and $M - u$ is a smaller bad filling pair. Hence this obstruction cannot occur, and M arises from a simplicial triangulation of B^3 .

It remains to check that this triangulation is clean. We have already seen that M is simplicial. We check the remaining conditions by the same minimality argument.

1. M is a pseudomanifold. For suppose some 2-simplex s occurs more than twice as the boundary of a 3-simplex of M . If $s \notin \sigma$, it must occur at least twice plus and twice minus; after removing any eligible tet it still occurs either twice plus or twice minus in $M - t$ in case (1), or in one or the other of M_1, M_2 in case (2), contradicting minimality. If $s \in \sigma$, it occurs at least twice plus in M : Remove some eligible tet that does not have s as a face to get a contradiction.
2. M has no edge $\{a, b\}$ whose link is disconnected. For the link is a 1-dimensional complex whose edges correspond to tets of M that contain $\{a, b\}$. The only way a component of the link can have fewer than three vertices is if it consists of a single edge $\{c, d\}$, corresponding to a tet $t = \{a, b, c, d\} \in M$. In this case $\{a, b\}$ must be an edge of σ , and $\{a, b, c\}, \{a, b, d\}$ the faces of σ that are adjacent along $\{a, b\}$. These are the only faces of σ that contain $\{a, b\}$, so no other component of the link can reduce to a single edge. If the link is disconnected, removing any eligible tet other than t will leave it disconnected, contradicting minimality.
3. M has no vertex whose link is disconnected: For if $\text{link}(\{a\}, M)$ is not connected, the only way removing a single tet t can render the link connected is if one component of the link has a single 2-simplex $\{b, c, d\}$ and $t = \{a, b, c, d\}$. In this case $\{a, b, c\}, \{a, b, d\}, \{a, c, d\}$ must all be faces of σ , making a a vertex of σ of degree 3, contradiction.

Hence M is clean. This contradicts the assumption that (σ, M) was bad, and proves the theorem. $\square$

Thanks to this theorem, we may take the liberty of saying that a taut filling of σ 'is' a simplicial triangulation τ of B^3 .

7 Shelling

We want to show that any taut filling τ of a triangulation σ of S^2 is *shellable*.

A *shelling* of a simplicial triangulation τ of B^3 is an ordering $(t_1, \dots, t_{|\tau|})$ of its tets such that gluing on one at a time maintains a triangulation of B^3 , i.e., so that for $k = 1, \dots, |\tau|$ the subcomplex τ_k generated by $\{t_1, \dots, t_k\}$ is a simplicial triangulation of B^3 . For $k > 1$ we distinguish the type of t_k according to the number of faces it shares with τ_{k-1} . In general the type can be any of 1, 2, 3, but in our application type 3 is ruled out because it would produce an interior vertex.

Call τ *freely shellable* if any $t \in \tau$ can serve as the initial tet of a shelling.

Theorem 3. *If σ is a simplicial triangulation of S^2 , any taut filling τ of σ is a freely shellable simplicial triangulation of B^3 .*

Proof. This is a corollary of the proof of Theorem 2.

Let $t_1 \in \tau$ be the designated initial tet of the shelling. Instead of thinking of how to build τ starting with t_1 , consider a recursive procedure—let’s call it ‘shucking’—for pulling τ apart so as to remove t_1 last.

The base case is when τ consists only of t_1 .

If τ has a vertex of degree 3 that belongs to a tet t other than t_1 , remove t and shuck $\tau - t$ ending with t_1 .

If τ has a vertex of degree 3 that belongs to t_1 , let t_2 be the tet of $\tau - t_1$ to which t_1 is attached; shuck $\tau - t_1$ ending with t_2 ; remove t_1 .

If there is no vertex of degree 3, we have at least $\maxdeg \geq 4$ disjointly eligible tets. Pick one such tet t that shares no face with t_1 . If $\tau - t$ doesn’t fall apart, remove t and shuck $\tau - t$ ending with t_1 . If $\tau - t$ falls apart into the almost disjoint union of τ_1 and τ_2 , assume $t \in \tau_1$; let t_2 be the tet where t attaches to τ_2 ; shuck τ_2 ending with t_2 ; remove t ; shuck τ_1 ending with t_1 .

□

8 Flag complex

A simplicial complex is a *flag complex* if every clique in its 1-skeleton occurs as a simplex. (Through gritted teeth we say that the complex ‘is flag’.) A simplicial triangulation σ of S^2 is flag just if it is *prime* (not the connected sum of smaller complexes) and not the boundary of a tetrahedron.

We can decompose any σ as the connected sum of prime components σ_i . (Of course, to reconstruct σ we need more than this, we need to know just how they are connected.) From the point of taut fillings these components are independent: Any taut filling τ of σ is a union of taut fillings τ_i .

Now we want to show that, whether or not σ itself is prime, any taut filling of σ is a flag complex.

Theorem 4. *If σ is a simplicial triangulation of S^2 , any taut filling τ of σ is a flag complex.*

Proof. The goal here is to show that none of these three taboo configurations occurs in τ :

1. The three edges of triangle $\{A, B, C\}$ without the 2-simplex ABC .
2. The six edges of the K_4 $\{A, B, C, D\}$ without the 3-simplex $ABCD$.
3. The ten edges of the K_5 $\{A, B, C, D, E\}$.

If we can rule out 1 and 2 then 3 will follow, because along with the K_5 we would have the five associated 3-simplices: Together these would form an S^3 , which is not possible as a subcomplex of the topological 3-ball τ .

To rule out 1 and 2, the strategy will be to consider a minimal counterexample, and find a modification that would produce a smaller counterexample.

If σ has a vertex of degree 3, it is a connected sum with one summand a single tetrahedron. We can pull off the vertex from σ , while removing the associated tet from τ , to produce a smaller counterexample.

If σ has no vertex of degree 3, there are at least $\maxdeg \geq 4$ disjointly eligible tet. Removing any one of these tets performs an ‘edge flip’ on the boundary σ , which either (a) remains a triangulation of the 2-sphere, or (b) becomes an almost disjoint union where two such triangulations are joined along an edge. We will show that some choice for the flip does not remove the taboo configuration. In case (a) we have a smaller counterexample already. In case (b) the key is that if a K_3 or K_4 occurs in the almost disjoint union, it occurs on one side or the other, also giving a counterexample.

Configuration 1: When we remove an eligible tet from τ to flip an edge of σ , the only edge that disappears from τ is the edge that gets flipped. There are at least four choices of the edge to flip, and three edges we must avoid flipping to preserve the taboo K_3 .

Configuration 2: Now there are six edges we must avoid flipping. Having dealt with configuration 1, we know that along with the six edges of $\{A, B, C, D\}$, τ must contain all four of the unoriented faces ABC , ABD , ACD , BCD . At most two of these can be boundary faces, because σ has no vertex of degree 3. So at most five of the K_5 edges are on the boundary. By treating the octahedron separately as a special case, we can assume $\maxdeg \geq 5$, so we have at least five boundary edges we can flip. The only potential problem is if five of the six K_4 edges are on the boundary. Say the interior edge is AD , so that the K_4 includes the faces ABC, BCD . If an eligible tet contains neither of these as one of its two boundary faces, we are fine. But at most two of our disjointly eligible tets can contain one of the faces ABC, BCD , leaving us at least $\maxdeg - 2$ other flipping options. $\square$

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